



COMP 2601 – Programming Part 3 (OODesign) Assignment 2 (100 points)

Goals

- Review unit testing, GUI, streams/filters

All class files should have the Java doc at the top, including your student name.

Name – student number

Submission:

ZIP file **Assignment2.zip** containing the project files.

Upload all the files zipped (.zip) to the Learning Hub under
Activities -> Assignments -> Assignment2

Submit just Zip files, any other types of archives will be rejected.

All submissions should be complete, should contain all files.

Coding Standards:

- Include your full name at the top of each file, using a Javadoc comment: for example: `/** @author Tiger Woods */`
- Do not use magic numbers in your code.
- Use all of the best practices we have used in class.

Please complete the following:

- Create a **Main class** with the main() method which shows a menu of options for the user to choose from. It shows the following screen and calls the chosen method as listed; then the menu appears again, repeat until the user types 3 to quit:

Menu

1. Countries and Capitals – displays a list of countries and capitals
 2. Countries Info – displays the info returned by the 4 functions of the CountryProcessor class.
 3. Exit – terminates the menu loop
- In a class called **CountryList**, displays all countries and their capitals in a JList (e.g. Canada: Ottawa), in alphabetical order by country name. Use the data from the **countries-and-capitals.txt** file. When the user closes the window, show the Main.main() menu.
 - In a class called **CountryProcessor**, create a function named processCountries() that calls all 4 functions. Use the data from the **countries-and-capitals.txt** file. Create a HashMap instance variable with country name as key, and its capital city name as value (e.g. "Canada": "Ottawa"). Use streams and filters to create the following functions which do exactly what they say. In all cases, before printing, collect the result into a java collection or local variable (e.g. List, String, Integer, etc...):
 1. printNumberOfCountries()
 2. printLongestCapitalCity()
 3. printShortestCountryName()
 4. printAllCountriesStartingWith(String substring)
 - **JUnit – create unit** tests to test the above 4 functions.
 - Create and upload (as pdf) the **UML Class diagram** showing the classes with their attributes, methods and relationships.

Main class

main() – shows a menu of 3 options: the menu repeats until the user choice is exit.

```
Assignment 2
Your Name
=====
1. Countries and Capitals
2. Countries Info
3. All Countries that start with
4. Exit
=====
Enter your choice [1-4]:
```

1. Option 1

CountryList class – display all countries and capitals in a **JList** in alphabetical order by CountryName. Use countries-and-capitals.txt

2. Option 2

CountryProcessor class – use countries-and-capitals.txt

processCountries() function calls 3 functions and consolidates the output in a nice, readable format for the user.

- HashMap instance country name (key)/capital city (value)
- Use streams and filters to create a collection and print it, for each of the functions:

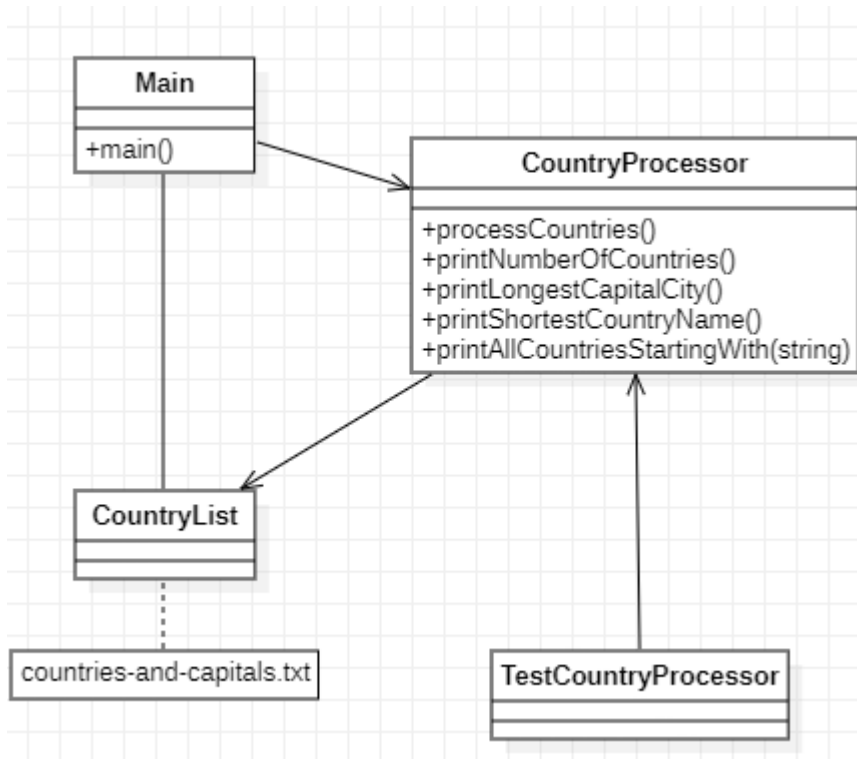
```
printNumberOfCountries()
printLongestCapitalCity()
printShortestCountryName()
```

3. Option 3

In same class implement the printAllCountriesStartingWith(String string) function. Ask the user to input the string.

Create Unit tests for the 3 methods called by processCountries().

Create the UML class diagram using Visual Paradigm Community Edition



Marking

Implement classes as expected	80	
▪ Main – menu – functional	10	
▪ CountryList class, JList	20	
▪ CountryProcessor class – 5 functions	40	
▪ Unit Tests	10	
UML class diagram	5	
Javadoc comments - complete for all classes and functions	10	
Coding style	5	
Total	100	